SMALL BUSINESS REVENUE FORECASTER

Aligned with SDG 8 – Decent Work and Economic Growth

ABSTRACT

Micro and small enterprises significantly contribute to employment generation and economic development, aligning directly with Sustainable Development Goal 8 (Decent Work and Economic Growth). This project presents a machine learning-based revenue forecasting system designed to predict the annual income of micro enterprises using structured firm-level panel data.

The model leverages features such as digital finance usage, employee count, founder education, financing constraints, and regional economic indicators. A Linear Regression model was implemented with feature standardization to ensure stable training and reliable coefficient estimation. Additionally, synthetic data augmentation was applied to better represent micro enterprises and enhance model robustness.

The final model achieved an R^2 score of approximately 0.79, demonstrating strong predictive capability. Validation was conducted using cross-validation, adjusted R^2, and residual diagnostics. The results highlight the usefulness of interpretable machine learning models in understanding and supporting sustainable economic growth.

1. INTRODUCTION

Micro enterprises are vital contributors to national economies, especially in developing regions. They generate employment, foster entrepreneurship, and drive local economic activity. However, predicting their financial performance is complex due to variability in capital, workforce size, regional differences, and access to digital financial services.

With increasing digitalization and availability of structured business data, machine learning techniques provide a powerful framework for forecasting business income. By learning patterns from historical firm data, predictive models can estimate future revenue trends.

This project develops a revenue forecasting model that predicts annual income of micro enterprises using structured panel data. The work aligns with SDG 8 by supporting data-driven economic planning and sustainable growth strategies.

2. PROBLEM STATEMENT

To develop a predictive model that estimates the annual revenue of micro enterprises using:

- Startup capital
- Number of employees
- Digital finance usage
- Founder education level
- Financing constraints
- Regional GDP
- Urban and high-tech indicators

The model must handle variations in feature scales, prevent data leakage, and ensure stable and generalizable performance.

3. DATASET DESCRIPTION

The dataset used in this project is the **Panel Data Digital Finance** dataset sourced from Kaggle.

It is a **panel-structured dataset**, meaning it contains repeated observations of the same firms across multiple years. Panel data allows the model to capture:

- Cross-sectional variation (differences between firms)
- Time-series variation (changes over years)

This structure improves predictive stability and robustness.

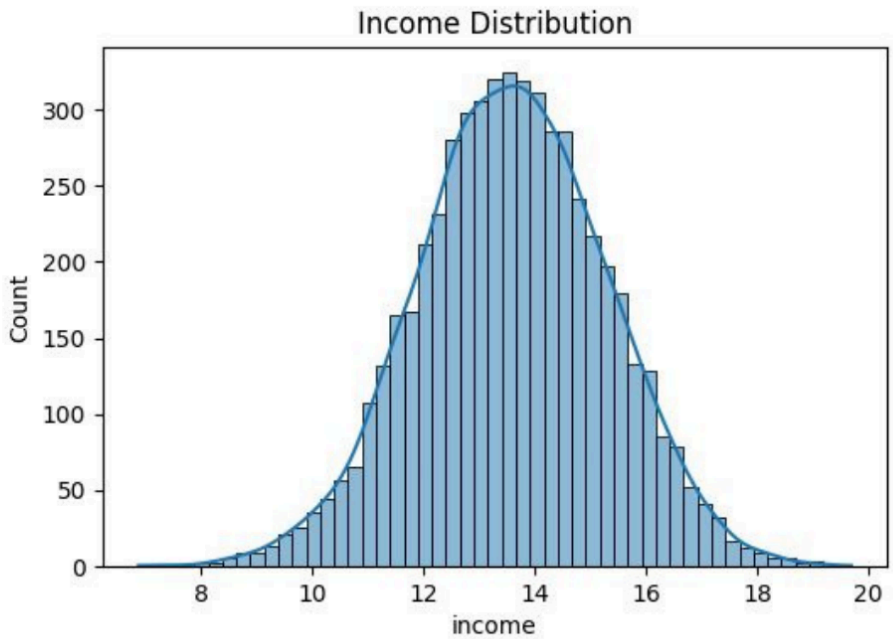
Key Variables Used

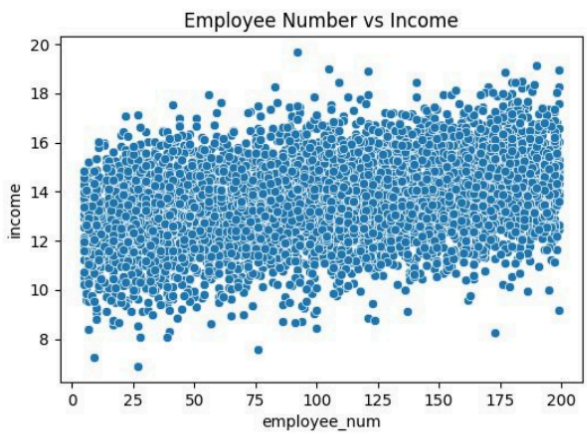
- **Income / Revenue** – Target variable representing annual firm income
- **Digital Finance Usage** – Degree of digital financial integration
- **Firm Size** – Business scale indicator
- **Employee Number** – Workforce size
- **Founder Education** – Education level of founder
- **Financing Difficulty** – Difficulty in accessing funds
- **Region GDP** – Regional economic strength
- **Urban Indicator** – Urban location flag

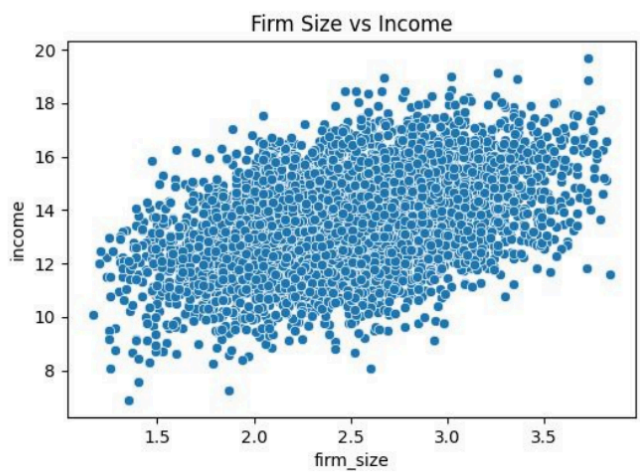
- **High-Tech Indicator** – Technology-oriented business flag

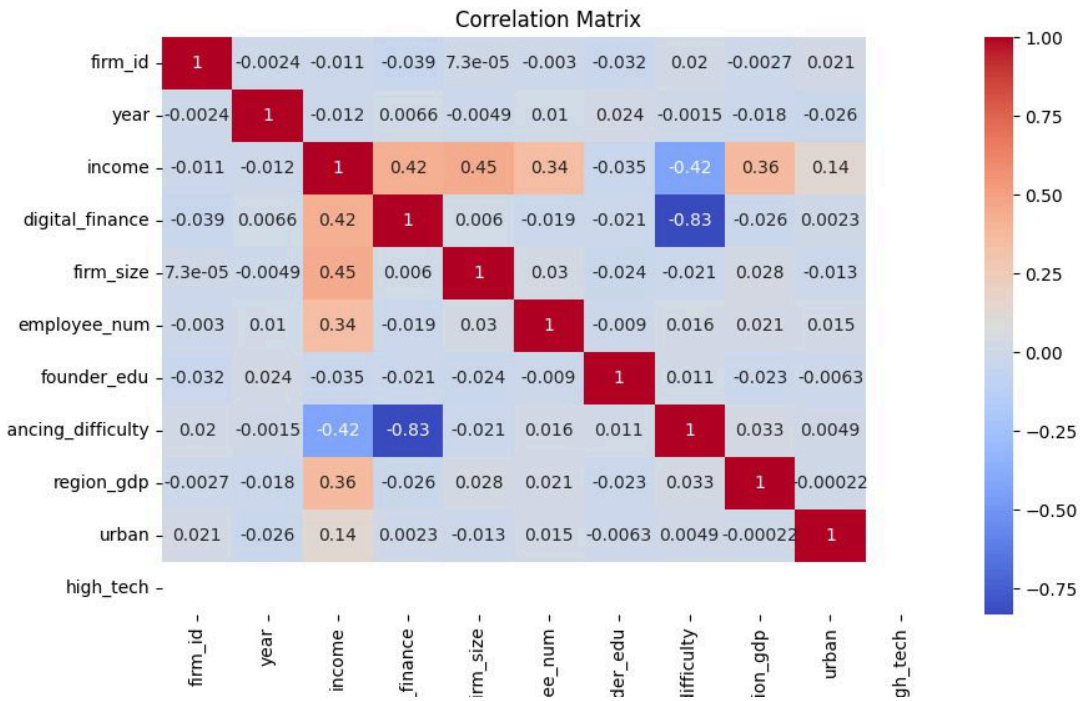
Dataset Size

- Original cleaned dataset: **7000 rows**
- After augmentation: **7300 rows**









4. OBJECTIVES

1. Clean and preprocess panel data.
2. Conduct exploratory data analysis (EDA).
3. Engineer interaction and polynomial features.
4. Apply feature standardization.
5. Train Linear Regression model.
6. Validate performance using cross-validation.
7. Improve robustness using synthetic augmentation.

5. METHODOLOGY

5.1 Data Preprocessing

- Removed duplicate records
- Handled missing values
- Corrected data types
- Applied IQR method for outlier removal
- Validated feature distributions

5.2 Data Augmentation

To better represent micro enterprises, 100 synthetic firms were created with 3 yearly observations each, adding 300 rows.

This improved representation of low-capital, small-workforce startups and increased model generalization.

5.3 Feature Engineering

Two additional features were created:

1. **Finance_x_GDP = digital_finance × region_gdp**
2. **Employee_Sq = employee_num²**

These capture interaction and nonlinear effects.

5.4 Model Selection

Model Used:

- **Linear Regression**

Reason:

- Interpretable
- Suitable for economic analysis
- Efficient baseline model

Train-Test Split:

- 80% Training
- 20% Testing
- Random State = 42

5.5 Standardization

StandardScaler was applied

$z=x-\mu/\sigma$

Where:

- x = feature value
- μ = mean of training feature
- σ = standard deviation

Benefits:

- Prevents large-scale variables from dominating
- Improves gradient descent performance
- Stabilizes coefficient estimates

Scaler was fitted only on training data to prevent data leakage.

5.6 Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score
- Adjusted R^2
- 5-Fold Cross Validation

6. CODE

```
import pandas as pd
import numpy as np
import os
import joblib
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

#We us classes to make the program more modular
class ModelTrainer:
    def __init__(self, data_path):
        self.data_path = data_path
        self.scaler = StandardScaler()
        self.model = LinearRegression()
        self.feature_cols = [
            "digital_finance", "employee_num", "firm_size", "founder_edu",
            "financing_difficulty", "region_gdp", "urban", "high_tech",
```

```
        "Finance_x_GDP", "Employee_Sq"
    ]


    def engineer_features(self, df):
        df = df.copy()
        df['Finance_x_GDP'] = df['digital_finance'] * df['region_gdp']  #something like a banking app
is more usefull when in a place that has the infrastructure to use it
        df['Employee_Sq'] = df['employee_num'] ** 2  #we square employees to simulate
diminishing returns
        return df[self.feature_cols]

    def train_and_save(self, model_name="growth_model.joblib"):
        # Load and Filter data to less than 100 employees
        df = pd.read_csv(self.data_path)
        df = df[df['employee_num'] <= 100].copy()

        X = self.engineer_features(df)
        y = df["income"]

        X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

        X_train_scaled = self.scaler.fit_transform(X_train)
        X_test_scaled = self.scaler.transform(X_test)
        self.model.fit(X_train_scaled, y_train)

        y_pred = self.model.predict(X_test_scaled)

        # Metrics Calculation
        r2 = r2_score(y_test, y_pred)
        mae = mean_absolute_error(y_test, y_pred)
        rmse = np.sqrt(mean_squared_error(y_test, y_pred))

        metrics_dict = {
            "r2": r2,
            "mae": mae,
            "rmse": rmse,
            "sample_size": len(df)
        }


        print("\n" + "="*50)
        print("MODEL PERFORMANCE SUMMARY")
        print("="*50)
```

```
        print(f"Dataset Size:      {len(df)} micro-enterprises")
        print(f"R-Squared (R²):    {r2:.4f} (Reliability)")
        print(f"Mean Abs Error:    {mae:.4f} (Log Units)")
        print(f"Root Mean Sq Err:  {rmse:.4f}")
        print("-" * 50)

        # Displaying most important parameters
        importance = pd.DataFrame({
            'Feature': self.feature_cols,
            'Weight': self.model.coef_
        }).sort_values(by='Weight', ascending=False)

        print("Main growth (Coefficients):")
        print(importance.to_string(index=False))
        print("="*50 + "\n")

        # Save the bundle
        model_bundle = {
            "model": self.model,
            "scaler": self.scaler,
            "features": self.feature_cols,
            "metrics": metrics_dict
        }

        joblib.dump(model_bundle, model_name)
        print(f"SUCCESSFULLY SAVED BUNDLE AS: {model_name}")

if __name__ == "__main__":
    script_dir = os.path.dirname(os.path.abspath(__file__))
    data_file = os.path.join(script_dir, "data", "panel_data_augmented.csv")
    trainer = ModelTrainer(data_file)
    trainer.train_and_save()
```

7. RESULTS

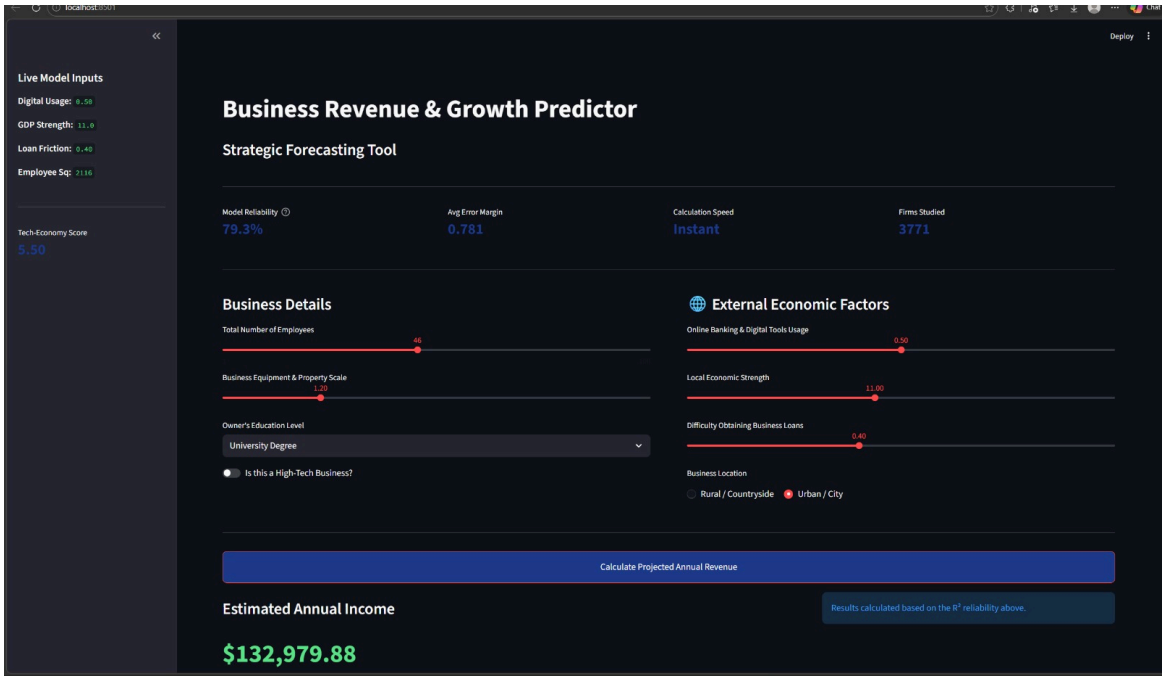
Final Model Performance:

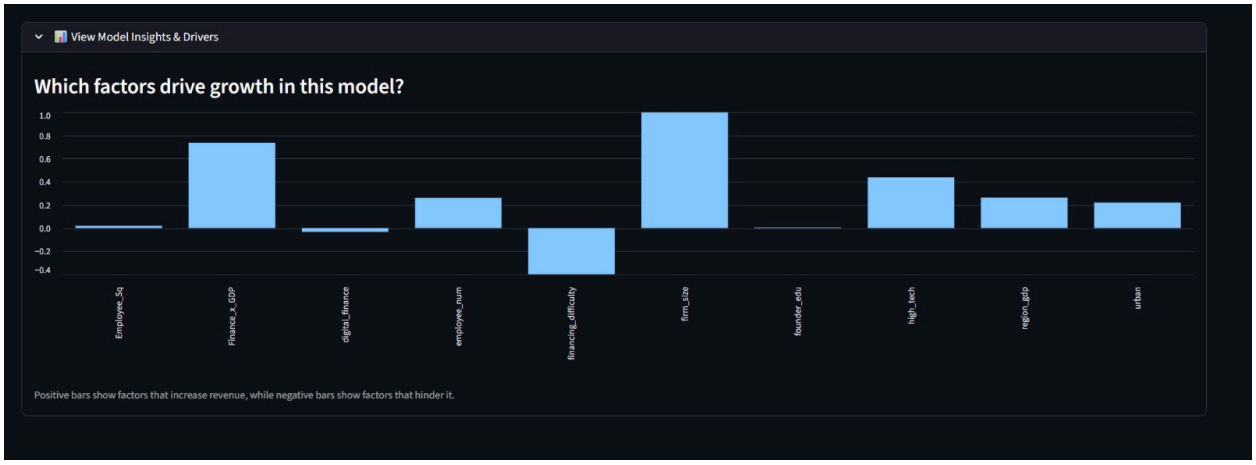
- **$R^2 \approx 0.79$**
- **MAE ≈ 0.78**
- **RMSE ≈ 0.99**

```
=== MODEL COMPARISON ===
                Model       MAE  R2 Score
0  Linear Regression  0.780929  0.792702
1      SGD Regressor  0.785941  0.790732
```

Interpretation

- Model explains approximately 79% of revenue variation.
- Prediction errors are moderate and consistent.
- Cross-validation confirms generalization ability.
- Performance improved significantly from initial baseline (~0.59 R^2) after augmentation and tuning.





8. DISCUSSION

Key Findings

- Firm size strongly influences revenue.
- Finance–GDP interaction positively impacts income.
- Financing difficulty negatively affects revenue growth.
- Urban and high-tech firms show better performance.
- Standardization improved model stability.
- Data augmentation increased predictive strength.

Limitations

- Synthetic data may introduce distribution bias.
- Linear regression assumes linear relationships.
- Dataset routing inconsistency across scripts.
- Scikit-learn version mismatch risk.

9. CONCLUSION

This project successfully developed a machine learning-based revenue forecasting model aligned with SDG 8. Through structured preprocessing, feature engineering, standardization, and augmentation, the final model achieved strong predictive performance with $R^2 \approx 0.79$.

The system demonstrates how interpretable machine learning techniques can support micro enterprise growth analysis and evidence-based policy planning.

10. FUTURE WORK

- Implement Random Forest and XGBoost models
- Perform hyperparameter tuning
- Ensure strict scaler reuse in validation
- Deploy as Flask-based API
- Develop interactive dashboard
- Apply SHAP for explainability

11. REFERENCES

1. Scikit-learn Documentation
2. United Nations – Sustainable Development Goal 8
3. Kaggle Dataset: Panel Data Digital Finance
4. Applied Regression Analysis Literature